

STATISTICAL TABLES

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Table ST1. Cumulative Binomial Probabilities, $\sum_{x=0}^r \binom{n}{x} p^x (1-p)^{n-x}$,
 $r = 0, 1, 2, \dots, n - 1$

<i>n</i>	<i>r</i>	<i>p</i>								
		0.01	0.05	0.10	0.20	0.25	0.30	0.333	0.40	0.50
2	0	0.9801	0.9025	0.8100	0.6400	0.5625	0.4900	0.4444	0.3600	0.2500
	1	0.9999	0.9975	0.9900	0.9600	0.9375	0.9100	0.8888	0.8400	0.7500
3	0	0.9703	0.8574	0.7290	0.5120	0.4219	0.3430	0.2963	0.2160	0.1250
	1	0.9997	0.9928	0.9720	0.8960	0.8438	0.7840	0.7407	0.6480	0.5000
	2	1.0000	0.9999	0.9990	0.9920	0.9844	0.9730	0.9629	0.9360	0.8750
4	0	0.9606	0.8145	0.6561	0.4096	0.3164	0.2401	0.1975	0.1296	0.0625
	1	0.9994	0.9860	0.9477	0.8192	0.7383	0.6517	0.5926	0.4742	0.3125
	2	1.0000	0.9995	0.9963	0.9728	0.9492	0.9163	0.8889	0.8198	0.6875
	3		1.0000	0.9999	0.9984	0.9961	0.9919	0.9877	0.9734	0.9375
5	0	0.9510	0.7738	0.5905	0.3277	0.2373	0.1681	0.1317	0.0778	0.0312
	1	0.9990	0.9774	0.9185	0.7373	0.6328	0.5283	0.4609	0.3370	0.1874
	2	1.0000	0.9988	0.9914	0.9421	0.8965	0.8370	0.7901	0.6826	0.4999
	3		0.9999	0.9995	0.9933	0.9844	0.9693	0.9547	0.9130	0.8124
	4		1.0000	1.0000	0.9997	0.9990	0.9977	0.9959	0.9898	0.9686
6	0	0.9415	0.7351	0.5314	0.2621	0.1780	0.1176	0.0878	0.0467	0.0156
	1	0.9986	0.9672	0.8857	0.6553	0.5340	0.4201	0.3512	0.2333	0.1094
	2	1.0000	0.9977	0.9841	0.9011	0.8306	0.7442	0.6804	0.5443	0.3438
	3		0.9998	0.9987	0.9830	0.9624	0.9294	0.8999	0.8208	0.6563
	4		0.9999	0.9999	0.9984	0.9954	0.9889	0.9822	0.9590	0.8907
	5		1.0000	1.0000	0.9999	0.9998	0.9991	0.9987	0.9959	0.9845
7	0	0.9321	0.6983	0.4783	0.2097	0.1335	0.0824	0.0585	0.0280	0.0078
	1	0.9980	0.9556	0.6554	0.5767	0.4450	0.3294	0.2633	0.1586	0.0625
	2	1.0000	0.9962	0.8503	0.8520	0.7565	0.6471	0.5706	0.4199	0.2266
	3		0.9998	0.9743	0.9667	0.9295	0.8740	0.8267	0.7102	0.5000
	4		1.0000	0.9973	0.9953	0.9872	0.9712	0.9547	0.9037	0.7734
	5			0.9998	0.9996	0.9987	0.9962	0.9931	0.9812	0.9375
	6			1.0000	1.0000	0.9999	0.9998	0.9995	0.9984	0.9922
8	0	0.9227	0.6634	0.4305	0.1678	0.1001	0.0576	0.0390	0.0168	0.0039
	1	0.9973	0.9427	0.8131	0.5033	0.3671	0.2553	0.1951	0.1064	0.0352
	2	0.9999	0.9942	0.9619	0.7969	0.6786	0.5518	0.4682	0.3154	0.1445
	3	1.0000	0.9996	0.9950	0.9437	0.8862	0.8059	0.7413	0.5941	0.3633
	4		1.0000	0.9996	0.9896	0.9727	0.9420	0.9120	0.8263	0.6367
	5			1.0000	0.9988	0.9958	0.9887	0.9803	0.9502	0.8555
	6				1.0000	0.9996	0.9987	0.9974	0.9915	0.9648
	7					1.0000	0.9999	0.9998	0.9993	0.9961
9	0	0.9135	0.6302	0.3874	0.1342	0.0751	0.0404	0.0260	0.0101	0.0020
	1	0.9965	0.9287	0.7748	0.4362	0.3004	0.1960	0.1431	0.0706	0.0196
	2	0.9999	0.9916	0.9470	0.7382	0.6007	0.4628	0.3772	0.2318	0.0899
	3	1.0000	0.9993	0.9916	0.9144	0.8343	0.7296	0.6503	0.4826	0.2540
	4		0.9999	0.9990	0.9805	0.9511	0.9011	0.8551	0.7334	0.5001

(Continued)

<i>n</i>	<i>r</i>	<i>p</i>								
		0.01	0.05	0.10	0.20	0.25	0.30	0.333	0.40	0.50
10	5		1.0000	0.9998	0.9970	0.9900	0.9746	0.9575	0.9006	0.7462
	6			0.9999	0.9998	0.9987	0.9956	0.9916	0.9749	0.9103
	7			1.0000	1.0000	0.9999	0.9995	0.9989	0.9961	0.9806
	8					1.0000	0.9999	0.9998	0.9996	0.9982
	0	0.9044	0.5987	0.3487	0.1074	0.0563	0.0282	0.0173	0.0060	0.0010
	1	0.9958	0.9138	0.7361	0.3758	0.2440	0.1493	0.1040	0.0463	0.0108
	2	1.0000	0.9884	0.9298	0.6778	0.5256	0.3828	0.2991	0.1672	0.0547
	3		0.9989	0.9872	0.8791	0.7759	0.6496	0.5592	0.3812	0.1719
	4		0.9999	0.9984	0.9672	0.9219	0.8497	0.7868	0.6320	0.3770
	5		1.0000	0.9999	0.9936	0.9803	0.9526	0.9234	0.8327	0.6231
11	6			1.0000	0.9991	0.9965	0.9894	0.9803	0.9442	0.8282
	7				0.9999	0.9996	0.9984	0.9966	0.9867	0.9454
	8				1.0000	1.0000	0.9998	0.9996	0.9973	0.9893
	9						1.0000	0.9999	0.9999	0.9991
	0	0.8954	0.5688	0.3138	0.0859	0.0422	0.0198	0.0116	0.0036	0.0005
	1	0.9948	0.8981	0.6974	0.3221	0.1971	0.1130	0.0752	0.0320	0.0059
	2	0.9998	0.9848	0.9104	0.6174	0.4552	0.3128	0.2341	0.1189	0.0327
	3	1.0000	0.9984	0.9815	0.8389	0.7133	0.5696	0.4726	0.2963	0.1133
	4		0.9999	0.9972	0.9496	0.8854	0.7897	0.7110	0.5328	0.2744
	5		1.0000	0.9997	0.9884	0.9657	0.9218	0.8779	0.7535	0.5000
12	6			1.0000	0.9981	0.9924	0.9784	0.9614	0.9007	0.7256
	7				0.9998	0.9988	0.9947	0.9912	0.9707	0.8867
	8				1.0000	0.9999	0.9994	0.9986	0.9941	0.9673
	9					1.0000	0.9999	0.9999	0.9993	0.9941
	10						1.0000	1.0000	1.0000	0.9995
	0	0.8864	0.5404	0.2824	0.0687	0.0317	0.0139	0.0077	0.0022	0.0002
	1	0.9938	0.8816	0.6590	0.2749	0.1584	0.0850	0.0540	0.0196	0.0032
	2	0.9998	0.9804	0.8892	0.5584	0.3907	0.2528	0.1811	0.0835	0.0193
	3	1.0000	0.9978	0.9744	0.7946	0.6488	0.4925	0.3931	0.2254	0.0730
	4	1.0000	0.9998	0.9957	0.9806	0.8424	0.7237	0.6315	0.4382	0.1939
13	5	1.0000	1.0000	0.9995	0.9961	0.9456	0.8822	0.8223	0.6652	0.3872
	6			1.0000	0.9994	0.9858	0.9614	0.9336	0.8418	0.6128
	7				0.9999	0.9972	0.9905	0.9812	0.9427	0.8062
	8				1.0000	0.9996	0.9983	0.9962	0.9848	0.9270
	9					10000	0.9998	0.9995	0.9972	0.9807
	10						1.0000	0.9999	0.9997	0.9968
	11							1.0000	1.0000	0.9998
	0	0.8775	0.5134	0.2542	0.0550	0.0238	0.0097	0.0052	0.0013	0.0000
	1	0.9928	0.8746	0.6214	0.2337	0.1267	0.0637	0.0386	0.0126	0.0017
	2	0.9997	0.9755	0.8661	0.5017	0.3326	0.2025	0.1388	0.0579	0.0112
	3	1.0000	0.9969	0.9659	0.7473	0.5843	0.4206	0.3224	0.1686	0.0462
	4		0.9997	0.9936	0.9009	0.7940	0.6543	0.5521	0.3531	0.1334
	5		1.0000	0.9991	0.9700	0.9198	0.8346	0.7587	0.5744	0.2905

Table ST1. (Continued)

<i>n</i>	<i>r</i>	<i>p</i>								
		0.01	0.05	0.10	0.20	0.25	0.30	0.333	0.40	0.50
14	6			0.9999	0.9930	0.9757	0.9376	0.8965	0.7712	0.5000
	7			1.0000	0.9988	0.9944	0.9818	0.9654	0.9024	0.7095
	8				0.9998	0.9990	0.9960	0.9912	0.9679	0.8666
	9				1.0000	0.9999	0.9994	0.9984	0.9922	0.9539
	10					1.0000	0.9999	0.9998	0.9987	0.9888
	11						1.0000	1.0000	0.9999	0.9983
	12								1.0000	0.9999
	0	0.8687	0.4877	0.2288	0.0440	0.0178	0.0068	0.0034	0.0008	0.0000
	1	0.9916	0.8470	0.5847	0.1979	0.1010	0.0475	0.0274	0.0081	0.0009
	2	0.9997	0.9700	0.8416	0.4480	0.2812	0.1608	0.1054	0.0398	0.0065
	3	1.0000	0.9958	0.9559	0.6982	0.5214	0.3552	0.2612	0.1243	0.0287
	4		0.9996	0.9908	0.8702	0.7416	0.5842	0.4755	0.2793	0.0898
15	5		1.0000	0.9986	0.9562	0.8884	0.7805	0.6898	0.4859	0.2120
	6			0.9998	0.9884	0.9618	0.9067	0.8506	0.6925	0.3953
	7			1.0000	0.9976	0.9897	0.9686	0.9424	0.8499	0.6048
	8				0.9996	0.9979	0.9917	0.9826	0.9417	0.7880
	9				1.0000	0.9997	0.9984	0.9960	0.9825	0.9102
	10					1.0000	0.9998	0.9993	0.9961	0.9713
	11						1.0000	0.9999	0.9994	0.9936
	12							1.0000	0.9999	0.9991
	13									0.9999
	0	0.8601	0.4633	0.2059	0.0352	0.0134	0.0048	0.0023	0.0005	0.0000
	1	0.9904	0.8291	0.5491	0.1672	0.0802	0.0353	0.0194	0.0052	0.0005
	2	0.9996	0.9638	0.8160	0.3980	0.2361	0.1268	0.0794	0.0271	0.0037
16	3	1.0000	0.9946	0.9444	0.6482	0.4613	0.2969	0.2092	0.0905	0.0176
	4		0.9994	0.9873	0.8358	0.6865	0.5255	0.4041	0.2173	0.0592
	5		1.0000	0.9978	0.9390	0.8516	0.7216	0.6184	0.4032	0.1509
	6			0.9997	0.9820	0.9434	0.8689	0.7970	0.6098	0.3036
	7			1.0000	0.9958	0.9827	0.9500	0.9118	0.7869	0.5000
	8				0.9992	0.9958	0.9848	0.9692	0.9050	0.6964
	9				0.9999	0.9992	0.9964	0.9915	0.9662	0.8491
	10				1.0000	0.9999	0.9993	0.9982	0.9907	0.9408
	11					1.0000	0.9999	0.9997	0.9981	0.9824
	12						1.0000	1.0000	0.9997	0.9963
	13								1.0000	0.9995
	14									1.0000

Source: For $n = 2$ through 10, adapted with permission from E. Parzen, *Modern Probability Theory and Its Applications*, John Wiley, New York, 1962. For $n = 11$ through 15, adapted with permission from *Tables of Cumulative Binomial Probability Distribution*, Harvard University Press, Cambridge, M.A., 1955.

Table ST2. Tail Probability Under Standard Normal Distribution^a

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2297	0.2266	0.2231	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1984	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010

Source: Adapted with permission from P. G. Hoel, *Introduction to Mathematical Statistics*, 4th ed., Wiley, New York, 1971, p. 391.

^aThis table gives the probability that the standard normal variable Z will exceed a given positive value z , that is, $P\{Z > z_\alpha\} = \alpha$. The probabilities for negative values of z are obtained by symmetry.

Table ST3. Critical Values Under Chi-Square Distribution^a

Degrees of Freedom	α													
	0.99	0.98	0.95	0.90	0.80	0.70	0.50	0.30	0.20	0.10	0.05	0.02	0.01	
1	0.000157	0.000628	0.00393	0.0158	0.0642	0.148	0.455	1.074	1.642	2.706	3.841	5.412	6.635	
2	0.0201	0.0404	0.103	0.211	0.446	0.713	1.386	2.408	3.219	4.605	5.991	7.824	9.210	
3	0.115	0.185	0.352	0.584	1.005	1.424	2.366	3.665	4.642	6.251	7.815	9.837	11.341	
4	0.297	0.429	0.711	1.064	1.649	2.195	3.357	4.878	5.989	7.779	9.488	11.668	13.277	
5	0.554	0.752	1.145	1.610	2.343	3.000	4.351	6.064	7.289	9.236	11.070	13.388	15.086	
6	0.872	1.134	1.635	2.204	3.070	3.828	5.348	7.231	8.558	10.645	12.592	15.033	16.812	
7	1.239	1.564	2.167	2.833	3.822	4.671	6.346	8.383	9.803	12.017	14.067	16.622	18.475	
8	1.646	2.032	2.733	3.490	4.594	5.527	7.344	9.524	11.030	13.362	15.507	18.168	20.090	
9	2.088	2.532	3.325	4.168	5.380	6.393	8.343	10.656	12.242	14.684	16.919	19.679	21.666	
10	2.558	3.059	3.940	4.865	6.179	7.267	9.342	11.781	13.442	15.987	18.307	21.161	23.209	
11	3.053	3.609	4.575	5.578	6.989	8.148	10.341	12.899	14.631	17.275	19.675	22.618	24.725	
12	3.571	4.178	5.226	6.304	7.807	9.034	11.340	14.011	15.812	18.549	21.026	24.054	26.217	
13	4.107	4.765	5.892	7.042	8.634	9.926	12.340	15.119	16.985	19.812	22.362	25.472	27.688	
14	4.660	5.368	6.571	7.790	9.467	10.821	13.339	16.222	18.151	21.064	23.685	26.873	29.141	
15	5.229	5.985	7.261	8.547	10.307	11.721	14.339	17.322	19.311	22.307	24.996	28.259	30.578	
16	5.812	6.614	7.962	9.312	11.152	12.624	15.338	18.418	20.465	23.542	26.296	29.633	32.000	
17	6.408	7.255	8.672	10.085	12.002	13.531	16.338	19.511	21.615	24.669	27.587	30.995	33.409	
18	7.015	7.906	9.390	10.865	12.857	14.440	17.338	20.601	22.760	25.989	28.869	32.346	34.805	
19	7.633	8.567	10.117	11.651	13.716	15.352	18.338	21.689	23.900	27.204	30.144	33.687	36.191	
20	8.260	9.237	10.851	12.443	14.578	16.266	19.337	22.775	25.038	28.412	31.410	35.020	37.566	
21	8.897	9.915	11.591	13.240	15.445	17.182	20.337	23.858	26.171	29.615	32.671	36.343	38.932	
22	9.542	10.600	12.338	14.041	16.314	18.101	21.337	24.939	27.301	30.813	33.924	37.659	40.289	
23	10.196	11.293	13.091	14.848	17.187	19.021	22.337	26.018	28.429	32.007	35.172	38.968	41.638	
24	10.856	11.992	13.848	15.659	18.062	19.943	23.337	27.096	29.553	33.196	36.415	40.270	42.980	
25	11.524	12.697	14.611	16.473	18.940	20.867	24.337	28.172	30.675	34.382	37.652	41.566	44.314	
26	12.198	13.409	15.379	17.292	19.820	21.792	25.336	29.246	31.795	35.563	38.885	42.856	45.642	
27	12.879	14.125	16.151	18.114	20.703	22.719	26.336	30.319	32.912	36.741	40.113	44.140	46.963	
28	13.565	14.847	16.928	18.939	21.588	23.647	27.336	31.391	34.027	37.916	41.337	45.419	48.278	
29	14.256	15.574	17.708	19.768	22.475	24.577	28.336	32.461	35.139	39.087	42.557	46.693	49.588	
30	14.953	16.306	18.493	20.599	23.364	25.508	29.336	33.530	36.250	40.256	43.773	47.962	50.892	

Source: Reproduced from *Statistical Methods for Research Workers*, 14th ed., 1972, with the permission of the Estate of R. A. Fisher, and Hafner Press.

^aFor degrees of freedom greater than 30, the expression $\sqrt{2\chi^2 - \sqrt{2n - 1}}$ may be used as a normal deviate with unit variance, where n is the number of degrees of freedom.

Table ST4. Student's *t*-Distribution^a

<i>n</i>	α				
	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Source: P. G. Hoel, *Introduction to Mathematical Statistics*, 4th ed., Wiley, New York, 1971, p. 393. Reprinted by permission of John Wiley & Sons, Inc.

^aThe first column lists the number of degrees of freedom (*n*). The headings of the other columns give probabilities (α) for *t* to exceed the entry value. Use symmetry for negative *t* values.

Table ST5. *F*-Distribution: 5% (Lightface Type) and 1% (Boldface Type) Points for the Distribution of *F*

Degrees of Freedom for Denominator (<i>n</i>)	Degrees of Freedom for Numerator (<i>m</i>)																								
	1	2	3	4	5	6	7	8	9	10	11	12	14	16	20	24	30	40	50	75	100	200	500	∞	
1	161	200	216	225	230	234	237	239	241	242	243	244	245	246	248	249	250	251	252	253	253	254	254	254	254
	4052	4999	5403	5625	5764	5859	5928	5981	6022	6056	6082	6106	6142	6169	6208	6234	6258	6286	6302	6323	6334	6352	6361	6366	
2	18.51	19.00	19.16	19.25	19.30	19.33	19.36	19.37	19.38	19.39	19.40	19.41	19.42	19.43	19.44	19.45	19.46	19.47	19.47	19.48	19.49	19.49	19.50	19.50	
	98.49	99.01	99.17	99.25	99.30	99.33	99.34	99.36	99.38	99.40	99.41	99.42	99.43	99.44	99.45	99.46	99.47	99.48	99.48	99.49	99.49	99.49	99.50	99.50	
3	10.13	9.55	9.28	9.12	9.01	8.94	8.88	8.84	8.81	8.78	8.76	8.74	8.71	8.69	8.66	8.64	8.62	8.60	8.58	8.57	8.56	8.54	8.54	8.53	
	34.12	30.81	29.46	28.71	28.24	27.91	27.67	27.49	27.34	27.23	27.13	27.05	26.92	26.83	26.69	26.60	26.50	26.41	26.30	26.27	26.23	26.18	26.14	26.12	
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.93	5.91	5.87	5.84	5.80	5.77	5.74	5.71	5.70	5.68	5.66	5.65	5.64	5.63	
	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.54	14.45	14.37	14.24	14.15	14.02	13.93	13.83	13.74	13.69	13.61	13.57	13.52	13.48	13.46	
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.78	4.74	4.70	4.68	4.64	4.60	4.56	4.53	4.50	4.46	4.44	4.42	4.40	4.38	4.37	4.36	
	16.26	13.27	12.06	11.39	10.97	10.67	10.45	10.27	10.15	10.05	9.96	9.89	9.77	9.68	9.55	9.47	9.38	9.29	9.24	9.17	9.13	9.07	9.04	9.02	
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.96	3.92	3.87	3.84	3.81	3.77	3.75	3.72	3.71	3.69	3.68	3.67	
	13.74	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79	7.72	7.60	7.52	7.39	7.31	7.23	7.14	7.09	7.02	6.99	6.94	6.90	6.88	
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.63	3.60	3.57	3.52	3.49	3.44	3.41	3.38	3.34	3.32	3.29	3.28	3.25	3.24	3.23	
	12.25	9.55	8.45	7.85	7.46	7.19	7.00	6.84	6.71	6.62	6.54	6.47	6.35	6.27	6.15	6.07	5.98	5.90	5.85	5.78	5.75	5.70	5.67	5.65	
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.34	3.31	3.28	3.23	3.20	3.15	3.12	3.08	3.05	3.03	3.00	2.98	2.96	2.94	2.93	
	11.26	8.65	7.59	7.01	6.63	6.37	6.19	6.03	5.91	5.82	5.74	5.67	5.56	5.48	5.36	5.28	5.20	5.11	5.06	5.00	4.96	4.91	4.88	4.86	
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.13	3.10	3.07	3.02	2.98	2.93	2.90	2.86	2.82	2.80	2.77	2.76	2.73	2.72	2.71	
	10.56	8.02	6.99	6.42	6.06	5.80	5.62	5.47	5.35	5.26	5.18	5.11	5.00	4.92	4.80	4.73	4.64	4.56	4.51	4.45	4.41	4.36	4.33	4.31	
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.97	2.94	2.91	2.86	2.82	2.77	2.74	2.70	2.67	2.64	2.61	2.59	2.56	2.55	2.54	
	10.04	7.56	6.55	5.99	5.64	5.39	5.21	5.06	4.95	4.85	4.78	4.71	4.60	4.52	4.41	4.33	4.25	4.17	4.12	4.05	4.01	3.96	3.93	3.91	
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.96	2.82	2.79	2.74	2.70	2.65	2.61	2.57	2.53	2.50	2.47	2.45	2.42	2.41	2.40	
	9.65	7.20	6.22	5.67	5.32	5.07	4.88	4.74	4.63	4.54	4.46	4.40	4.29	4.21	4.10	4.02	3.94	3.86	3.80	3.74	3.70	3.66	3.62	3.60	
12	4.75	3.88	3.49	3.26	3.11	3.00	2.92	2.85	2.80	2.76	2.72	2.69	2.64	2.60	2.54	2.50	2.46	2.42	2.40	2.36	2.35	2.32	2.31	2.30	
	9.33	6.93	5.95	5.41	5.06	4.82	4.65	4.50	4.39	4.30	4.22	4.16	4.05	3.98	3.86	3.78	3.70	3.61	3.56	3.49	3.46	3.41	3.38	3.36	
13	4.67	3.80	3.41	3.18	3.02	2.92	2.84	2.77	2.72	2.67	2.63	2.60	2.55	2.51	2.46	2.42	2.38	2.34	2.32	2.28	2.26	2.24	2.22	2.21	
	9.07	6.70	5.74	5.20	4.86	4.62	4.44	4.30	4.19	4.10	4.02	3.96	3.85	3.78	3.67	3.59	3.51	3.42	3.37	3.30	3.27	3.21	3.18	3.16	

14	4.60	3.74	3.34	3.11	2.96	2.85	2.77	2.70	2.65	2.60	2.56	2.53	2.48	2.44	2.39	2.35	2.31	2.27	2.24	2.21	2.19	2.16	2.14	2.13
	8.86	6.51	5.56	5.03	4.69	4.46	4.28	4.14	4.03	3.94	3.86	3.80	3.70	3.62	3.51	3.43	3.34	3.26	3.21	3.14	3.11	3.06	3.02	3.00
15	4.54	3.68	3.29	3.06	2.90	2.79	2.70	2.64	2.59	2.55	2.51	2.48	2.43	2.39	2.33	2.29	2.25	2.21	2.18	2.15	2.12	2.10	2.08	2.07
	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.73	3.67	3.56	3.48	3.36	3.29	3.20	3.12	3.07	3.00	2.97	2.92	2.89	2.87
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.45	2.42	2.37	2.33	2.28	2.24	2.20	2.16	2.13	2.09	2.07	2.04	2.02	2.01
	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.61	3.55	3.45	3.37	3.25	3.18	3.10	3.01	2.96	2.89	2.86	2.80	2.77	2.75
17	4.45	3.59	3.20	2.96	2.81	2.70	2.62	2.55	2.50	2.45	2.41	2.38	2.33	2.29	2.23	2.19	2.15	2.11	2.08	2.04	2.02	1.99	1.97	1.96
	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.52	3.45	3.35	3.27	3.16	3.08	3.00	2.92	2.86	2.79	2.76	2.70	2.67	2.65
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34	2.29	2.25	2.19	2.15	2.11	2.07	2.04	2.00	1.98	1.95	1.93	1.92
	8.28	6.01	5.09	4.58	4.25	4.01	3.85	3.71	3.60	3.51	3.44	3.37	3.27	3.19	3.07	3.00	2.91	2.83	2.78	2.71	2.68	2.62	2.59	2.57
19	4.38	3.52	3.13	2.90	2.74	2.63	2.55	2.48	2.43	2.38	2.34	2.31	2.26	2.21	2.15	2.11	2.07	2.02	2.00	1.96	1.94	1.91	1.90	1.88
	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.36	3.30	3.19	3.12	3.00	2.92	2.84	2.76	2.70	2.63	2.60	2.54	2.51	2.49
20	4.35	3.49	3.10	2.87	2.71	2.60	2.52	2.45	2.40	2.35	2.31	2.28	2.23	2.18	2.12	2.08	2.04	1.99	1.96	1.92	1.90	1.87	1.85	1.84
	8.10	5.85	4.94	4.43	4.10	3.87	3.71	3.56	3.45	3.37	3.30	3.23	3.13	3.05	2.94	2.86	2.77	2.69	2.63	2.56	2.53	2.47	2.44	2.42
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.28	2.25	2.20	2.15	2.09	2.05	2.00	1.96	1.93	1.89	1.87	1.84	1.82	1.81
	8.02	5.78	4.87	4.37	4.04	3.81	3.65	3.51	3.40	3.31	3.24	3.17	3.07	2.99	2.88	2.80	2.72	2.63	2.58	2.51	2.47	2.42	2.38	2.36
22	4.30	3.44	3.05	2.82	2.66	2.55	2.47	2.40	2.35	2.30	2.26	2.23	2.18	2.13	2.07	2.03	1.98	1.93	1.91	1.87	1.84	1.81	1.80	1.78
	7.94	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18	3.12	3.02	2.94	2.83	2.75	2.67	2.58	2.53	2.46	2.42	2.37	2.33	2.31
23	4.28	3.42	3.03	2.80	2.64	2.53	2.45	2.38	2.32	2.28	2.24	2.20	2.14	2.10	2.04	2.00	1.96	1.91	1.88	1.84	1.82	1.79	1.77	1.76
	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.14	3.07	2.97	2.89	2.78	2.70	2.62	2.53	2.48	2.41	2.37	2.32	2.28	2.26
24	4.26	3.40	3.01	2.78	2.62	2.51	2.43	2.36	2.30	2.26	2.22	2.18	2.13	2.09	2.02	1.98	1.94	1.89	1.86	1.82	1.80	1.76	1.74	1.73
	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.25	3.17	3.09	3.03	2.93	2.85	2.74	2.66	2.58	2.49	2.44	2.36	2.33	2.27	2.23	2.21
25	4.24	3.38	2.99	2.76	2.60	2.49	2.41	2.34	2.28	2.24	2.20	2.16	2.11	2.06	2.00	1.96	1.92	1.87	1.84	1.80	1.77	1.74	1.72	1.71
	7.77	5.57	4.68	4.18	3.86	3.63	3.46	3.32	3.21	3.13	3.05	2.99	2.89	2.81	2.70	2.62	2.54	2.45	2.40	2.32	2.29	2.23	2.19	2.17
26	4.22	3.37	2.89	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15	2.10	2.05	1.99	1.95	1.90	1.85	1.82	1.78	1.76	1.72	1.70	1.69
	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.17	3.09	3.02	2.96	2.86	2.77	2.66	2.58	2.50	2.41	2.36	2.28	2.25	2.19	2.15	2.13

Table ST5. (Continued)

Degrees of Freedom for Denominator (<i>n</i>)	Degrees of Freedom for Numerator, <i>m</i>																		
	1	2	3	4	5	6	7	8	9	10	11	12	14	16	20	24	30	40	50
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.30	2.25	2.20	2.16	2.13	2.08	2.03	1.97	1.93	1.88	1.84	1.80
	7.68	5.49	4.60	4.11	3.79	3.56	3.39	3.26	3.14	3.06	2.98	2.93	2.83	2.74	2.63	2.55	2.47	2.38	2.23
28	4.20	3.34	2.95	2.71	2.56	2.44	2.36	2.29	2.24	2.19	2.15	2.12	2.06	2.02	1.96	1.91	1.87	1.81	1.78
	7.64	5.45	4.57	4.07	3.76	3.53	3.36	3.23	3.11	3.03	2.95	2.90	2.80	2.71	2.60	2.52	2.44	2.35	2.30
29	4.18	3.33	2.93	2.70	2.54	2.43	2.35	2.28	2.22	2.18	2.14	2.10	2.05	2.00	1.94	1.90	1.85	1.80	1.77
	7.60	5.52	4.54	4.04	3.73	3.50	3.33	3.20	3.08	3.00	2.92	2.87	2.77	2.68	2.57	2.49	2.41	2.32	2.27
30	4.17	3.32	2.92	2.69	2.53	2.42	2.34	2.27	2.21	2.16	2.12	2.09	2.04	1.99	1.93	1.89	1.84	1.79	1.76
	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.06	2.98	2.90	2.84	2.74	2.66	2.55	2.47	2.38	2.29	2.24
32	4.15	3.30	2.90	2.67	2.51	2.40	2.32	2.25	2.19	2.14	2.10	2.07	2.02	1.97	1.91	1.86	1.82	1.76	1.74
	7.50	5.34	4.46	3.97	3.66	3.42	3.25	3.12	3.01	2.94	2.86	2.80	2.70	2.62	2.51	2.42	2.34	2.25	2.20
34	4.13	3.28	2.88	2.65	2.49	2.38	2.30	2.23	2.17	2.12	2.08	2.05	2.00	1.95	1.89	1.84	1.80	1.74	1.71
	7.44	5.29	4.42	3.93	3.61	3.38	3.21	3.08	2.97	2.89	2.82	2.76	2.66	2.58	2.47	2.38	2.30	2.21	2.15
36	4.11	3.26	2.86	2.63	2.48	2.36	2.28	2.21	2.15	2.10	2.06	2.03	1.89	1.93	1.87	1.82	1.78	1.72	1.69
	7.39	5.25	4.38	3.89	3.58	3.35	3.18	3.04	2.94	2.86	2.78	2.72	2.62	2.54	2.43	2.35	2.26	2.17	2.12
38	4.10	3.25	2.85	2.62	2.46	2.35	2.26	2.19	2.14	2.09	2.05	2.02	1.96	1.92	1.85	1.80	1.76	1.71	1.67
	7.35	5.21	4.34	3.86	3.54	3.32	3.15	3.02	2.91	2.82	2.75	2.69	2.59	2.51	2.30	2.32	2.22	2.14	2.08
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.07	2.04	2.00	1.95	1.90	1.84	1.79	1.74	1.69	1.66
	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.88	2.80	2.73	2.66	2.56	2.49	2.37	2.29	2.20	2.11	2.05
42	4.07	3.22	2.83	2.59	2.44	2.32	2.24	2.17	2.11	2.06	2.02	1.99	1.94	1.89	1.82	1.78	1.73	1.68	1.64
	7.27	5.15	4.29	3.80	3.49	3.26	3.10	2.96	2.86	2.77	2.70	2.64	2.54	2.46	2.35	2.26	2.17	2.08	2.02
44	4.06	3.21	2.82	2.58	2.43	2.31	2.23	2.16	2.10	2.05	2.01	1.98	1.92	1.88	1.81	1.76	1.72	1.66	1.63
	7.24	5.12	4.26	3.78	3.46	3.24	3.07	2.94	2.84	2.75	2.68	2.62	2.52	2.44	2.32	2.24	2.15	2.06	2.00
46	4.05	3.20	2.81	2.57	2.42	2.30	2.22	2.14	2.09	2.04	2.00	1.97	1.91	1.87	1.80	1.75	1.71	1.65	1.62
	7.21	5.10	4.24	3.76	3.44	3.22	3.05	2.92	2.82	2.73	2.66	2.60	2.50	2.42	2.40	2.22	2.13	2.04	1.98
48	4.04	3.19	2.80	2.56	2.41	2.30	2.21	2.14	2.08	2.03	1.99	1.96	1.90	1.86	1.79	1.74	1.70	1.64	1.61
	7.19	5.08	4.22	3.74	3.42	3.20	3.04	2.90	2.80	2.71	2.64	2.58	2.48	2.40	2.28	2.20	2.11	2.02	1.96

50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.02	1.98	1.95	1.90	1.85	1.78	1.74	1.69	1.63	1.60	1.55	1.52	1.48	1.46	1.44
	7.17	5.06	4.20	3.72	3.41	3.18	3.02	2.88	2.78	2.70	2.62	2.56	2.46	2.39	2.26	2.18	2.10	2.00	1.94	1.86	1.82	1.76	1.71	1.68
55	4.02	3.17	2.78	2.54	2.38	2.27	2.18	2.11	2.05	2.00	1.97	1.93	1.88	1.83	1.76	1.72	1.67	1.61	1.58	1.52	1.50	1.46	1.43	1.41
	7.12	5.01	4.16	3.68	3.37	3.15	2.98	2.85	2.75	2.66	2.59	2.53	2.43	2.35	2.23	2.15	2.06	1.96	1.90	1.82	1.78	1.71	1.66	1.64
60	4.00	3.15	2.76	2.52	2.37	2.25	2.17	2.10	2.04	1.99	1.95	1.92	1.86	1.81	1.75	1.70	1.65	1.59	1.56	1.50	1.48	1.44	1.41	1.39
	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56	2.50	2.40	2.32	2.20	2.12	2.03	1.93	1.87	1.79	1.74	1.68	1.63	1.60
65	3.99	3.14	2.75	2.51	2.36	2.24	2.15	2.08	2.02	1.98	1.94	1.90	1.85	1.80	1.73	1.68	1.63	1.57	1.54	1.49	1.46	1.42	1.39	1.37
	7.04	4.95	4.10	3.62	3.31	3.09	2.93	2.79	2.70	2.61	2.54	2.47	2.30	2.23	2.18	2.09	2.00	1.90	1.84	1.76	1.71	1.64	1.60	1.56
70	3.98	3.13	2.74	2.50	2.35	2.32	2.14	2.07	2.01	1.97	1.93	1.89	1.84	1.79	1.72	1.67	1.62	1.56	1.53	1.47	1.45	1.40	1.37	1.35
	7.01	4.92	4.08	3.60	3.29	3.07	2.91	2.77	2.67	2.59	2.51	2.45	2.35	2.28	2.15	2.07	1.98	1.88	1.82	1.74	1.69	1.63	1.56	1.53
80	3.96	3.11	2.72	2.48	2.33	2.21	2.12	2.05	1.99	1.95	1.91	1.88	1.82	1.77	1.70	1.65	1.60	1.54	1.51	1.45	1.42	1.38	1.35	1.32
	6.96	4.88	4.04	3.56	3.25	3.04	2.87	2.74	2.64	2.55	2.48	2.41	2.32	2.24	2.11	2.03	1.94	1.84	1.78	1.70	1.65	1.57	1.52	1.49
100	3.94	3.09	2.70	2.46	2.30	2.19	2.10	2.03	1.97	1.92	1.88	1.85	1.79	1.75	1.68	1.63	1.57	1.51	1.48	1.42	1.39	1.34	1.30	1.28
	6.90	4.82	3.98	3.51	3.20	2.99	2.82	2.69	2.59	2.51	2.43	2.36	2.26	2.19	2.06	1.98	1.89	1.79	1.73	1.64	1.59	1.51	1.46	1.43
125	3.92	3.07	2.68	2.44	2.29	2.17	2.08	2.01	1.95	1.90	1.86	1.83	1.77	1.72	1.65	1.60	1.55	1.49	1.45	1.39	1.36	1.31	1.27	1.25
	6.84	4.78	3.94	3.47	3.17	2.95	2.79	2.65	2.56	2.47	2.40	2.33	2.23	2.15	2.03	1.94	1.85	1.75	1.68	1.59	1.54	1.46	1.40	1.37
150	3.91	3.06	2.67	2.43	2.27	2.16	2.07	2.00	1.94	1.89	1.85	1.82	1.76	1.71	1.64	1.59	1.54	1.47	1.44	1.37	1.34	1.29	1.25	1.22
	6.81	4.75	3.91	3.44	3.13	2.92	2.76	2.62	2.53	2.44	2.37	2.30	2.20	2.12	2.00	1.91	1.83	1.72	1.66	1.56	1.51	1.43	1.37	1.33
200	3.89	3.04	2.65	2.41	2.26	2.14	2.05	1.98	1.92	1.87	1.83	1.80	1.74	1.69	1.62	0.157	1.52	1.45	1.42	1.35	1.32	1.26	1.22	1.19
	6.76	4.71	3.88	3.41	3.11	2.90	2.73	2.60	2.50	2.41	2.34	2.28	1.17	2.09	1.97	1.88	1.79	1.69	1.62	1.53	1.48	1.39	1.33	1.28
400	3.86	3.02	2.62	2.39	2.23	2.12	2.03	1.96	1.90	1.85	1.81	1.78	1.72	1.67	1.60	1.54	1.49	1.42	1.38	1.32	1.28	1.22	1.16	1.13
	6.70	4.66	3.83	3.36	3.06	2.85	2.69	2.55	2.46	2.37	2.29	2.23	2.12	2.04	1.92	1.84	1.74	1.64	1.57	1.47	1.42	1.32	1.24	1.19
1000	3.85	3.00	2.61	2.38	2.22	2.10	2.02	1.95	1.89	1.84	1.80	1.76	1.70	1.65	1.58	1.53	1.47	1.41	1.36	1.30	1.26	1.19	1.13	1.08
	6.66	4.62	3.80	3.34	3.04	2.82	2.66	2.53	2.43	2.34	2.26	2.20	2.09	2.01	1.89	1.81	1.71	1.61	1.54	1.44	1.38	1.28	1.19	1.11
∞	3.84	2.99	2.60	2.37	2.21	2.09	2.01	1.94	1.88	1.83	1.79	1.75	1.69	1.64	1.57	1.52	1.46	1.40	1.35	1.28	1.24	1.17	1.11	1.00
	6.64	4.60	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.24	2.18	2.07	1.99	1.87	1.79	1.69	1.59	1.52	1.41	1.36	1.25	1.15	1.00

Source: Reprinted by permission from George W. Snedecor and William G. Cochran, *Statistical Methods*, 6th ed., © 1967 by Iowa State University Press, Ames, I.A.

Table ST6. Random Normal Numbers, $\mu = 0$ and $\sigma = 1$

1	2	3	4	5	6	7	8	9	10
0.464	0.137	2.455	-0.323	-0.068	0.290	-0.288	1.298	0.241	-0.957
0.060	-2.526	-0.531	-0.194	0.543	-1.558	0.187	-1.190	0.022	0.525
1.486	-0.354	-0.634	0.697	0.926	1.375	0.785	-0.963	-0.853	-1.865
1.022	-0.472	1.279	3.521	0.571	-1.851	0.194	1.192	-0.501	-0.273
1.394	-0.555	0.046	0.321	2.945	1.974	-0.258	0.412	0.439	-0.035
0.906	-0.513	-0.525	0.595	0.881	-0.934	1.579	0.161	-1.885	0.371
1.179	-1.055	0.007	0.769	0.971	0.712	1.090	-0.631	-0.255	-0.702
-1.501	-0.488	-0.162	-0.136	1.033	0.203	0.448	0.748	-0.423	-0.432
-0.690	0.756	-1.618	-0.345	-0.511	-2.051	-0.457	-0.218	0.857	-0.465
1.372	0.225	0.378	0.761	0.181	-0.736	0.960	-1.530	-0.260	0.120
-0.482	1.678	-0.057	-1.229	-0.486	0.856	-0.491	-1.983	-2.830	-0.238
-1.376	-0.150	1.356	-0.561	-0.256	-0.212	0.219	0.779	0.953	-0.869
-1.010	0.598	-0.918	1.598	0.065	0.415	-0.169	0.313	-0.973	-1.016
-0.005	-0.899	0.012	-0.725	1.147	-0.121	1.096	0.481	-1.691	0.417
1.393	1.163	-0.911	1.231	-0.199	-0.246	1.239	-2.574	-0.558	0.056
-1.787	-0.261	1.237	1.046	-0.508	-1.630	-0.146	-0.392	-0.627	0.561
-0.105	-0.357	-1.384	0.360	-0.992	-0.116	-1.698	-2.832	-1.108	-2.357
-1.339	1.827	-0.959	0.424	0.969	-1.141	-1.041	0.362	-1.726	1.956
1.041	0.535	0.731	1.377	0.983	-1.330	1.620	-1.040	0.524	-0.281
0.279	-2.056	0.717	-0.873	-1.096	-1.396	1.047	0.089	-0.573	0.932
-1.805	-2.008	-1.633	0.542	0.250	-0.166	0.032	0.079	0.471	-1.029
-1.186	1.180	1.114	0.882	1.265	-0.202	0.151	-0.376	-0.310	0.479
0.658	-1.141	1.151	-1.210	0.927	0.425	0.290	-0.902	0.610	2.709
-0.439	0.358	-1.939	0.891	-0.227	0.602	0.873	-0.437	-0.220	-0.057
-1.399	-0.230	0.385	-0.649	-0.577	0.237	-0.289	0.513	0.738	-0.300
0.199	0.208	-1.083	-0.219	-0.291	1.221	1.119	0.004	-2.015	-0.594
0.159	0.272	-0.313	0.084	-2.828	-0.430	-0.792	-1.275	-0.623	-1.047
2.273	0.606	0.606	-0.747	0.247	1.291	0.063	-1.793	-0.699	-1.347
0.041	-0.307	0.121	0.790	-0.584	0.541	0.484	-0.986	0.481	0.996
-1.132	-2.098	0.921	0.145	0.446	-1.661	1.045	-1.363	-0.586	-1.023
0.768	0.079	-1.473	0.034	-2.127	0.665	0.084	-0.880	-0.579	0.551
0.375	-1.658	-0.851	0.234	-0.656	0.340	-0.086	-0.158	-0.120	0.418
-0.513	-0.344	0.210	-0.736	1.041	0.008	0.427	-0.831	0.191	0.074
0.292	-0.521	1.266	-1.206	-0.899	0.110	-0.528	-0.813	0.071	0.524
1.026	2.990	-0.574	-0.491	-1.114	1.297	-1.433	-1.345	-3.001	0.479
-1.334	1.278	-0.568	-0.109	-0.515	-0.566	2.923	0.500	0.359	0.326
-0.287	-0.144	-0.254	0.574	-0.451	-1.181	-1.190	-0.318	-0.094	1.114
0.161	-0.886	-0.921	-0.509	1.410	-0.518	0.192	-0.432	1.501	1.068
-1.346	0.193	-1.202	0.394	-1.045	0.843	0.942	1.045	0.031	0.772
1.250	-0.199	-0.288	1.810	1.378	0.584	1.216	0.733	0.402	0.226
0.630	-0.537	0.782	0.060	0.499	-0.431	1.705	1.164	0.884	-0.298
0.375	-1.941	0.247	-0.491	0.665	-0.135	-0.145	-0.498	0.457	1.064

(Continued)

1	2	3	4	5	6	7	8	9	10
−1.420	0.489	−1.711	−1.186	0.754	−0.732	−0.066	1.006	−0.798	0.162
−0.151	−0.243	−0.430	−0.762	0.298	1.049	1.810	2.885	−0.768	−0.129
−0.309	0.531	0.416	−1.541	1.456	2.040	−0.124	0.196	0.023	−1.204
0.424	−0.444	0.593	0.993	−0.106	0.116	0.484	−1.272	1.066	1.097
0.593	0.658	−1.127	−1.407	−1.579	−1.616	1.458	1.262	0.736	−0.916
0.862	−0.885	−0.142	−0.504	0.532	1.381	0.022	−0.281	−0.342	1.222
0.235	−0.628	−0.023	−0.463	−0.899	−0.394	−0.538	1.707	−0.188	−1.153
−0.853	0.402	0.777	0.833	0.410	−0.349	−1.094	0.580	1.395	1.298

Source: From tables of the RAND Corporation, by permission.

Table ST7. Critical Values of the Kolmogorov–Smirnov One-Sample Test Statistic^a

One-Sided Test:											
$\alpha =$	0.10	0.05	0.025	0.01	0.005	$\alpha =$	0.10	0.05	0.025	0.01	0.005
Two-Sided Test:											
$\alpha =$	0.20	0.10	0.05	0.02	0.01	$\alpha =$	0.20	0.10	0.05	0.02	0.01
$n = 1$	0.900	0.950	0.975	0.990	0.995	$n = 21$	0.226	0.259	0.287	0.321	0.344
2	0.684	0.776	0.842	0.900	0.929	22	0.221	0.253	0.281	0.314	0.337
3	0.565	0.636	0.708	0.785	0.829	23	0.216	0.247	0.275	0.307	0.330
4	0.493	0.565	0.624	0.689	0.734	24	0.212	0.242	0.269	0.301	0.323
5	0.447	0.509	0.563	0.627	0.669	25	0.208	0.238	0.264	0.295	0.317
6	0.410	0.468	0.519	0.577	0.617	26	0.204	0.233	0.259	0.290	0.311
7	0.381	0.436	0.483	0.538	0.576	27	0.200	0.229	0.254	0.284	0.305
8	0.358	0.410	0.454	0.507	0.542	28	0.197	0.225	0.250	0.279	0.300
9	0.339	0.387	0.430	0.480	0.513	29	0.193	0.221	0.246	0.275	0.295
10	0.323	0.369	0.409	0.457	0.489	30	0.190	0.218	0.242	0.270	0.290
11	0.308	0.352	0.391	0.437	0.468	31	0.187	0.214	0.238	0.266	0.285
12	0.296	0.338	0.375	0.419	0.449	32	0.184	0.211	0.234	0.262	0.281
13	0.285	0.325	0.361	0.404	0.432	33	0.182	0.208	0.231	0.258	0.277
14	0.275	0.314	0.349	0.390	0.418	34	0.179	0.205	0.227	0.254	0.273
15	0.266	0.304	0.338	0.377	0.404	35	0.177	0.202	0.224	0.251	0.269
16	0.258	0.295	0.327	0.366	0.392	36	0.174	0.199	0.221	0.247	0.265
17	0.250	0.286	0.318	0.355	0.381	37	0.172	0.196	0.218	0.244	0.262
18	0.244	0.279	0.309	0.346	0.371	38	0.170	0.194	0.215	0.241	0.258
19	0.237	0.271	0.301	0.337	0.361	39	0.168	0.191	0.213	0.238	0.255
20	0.232	0.265	0.294	0.329	0.352	40	0.165	0.189	0.210	0.235	0.252
Approximation for $n > 40$							$\frac{1.07}{\sqrt{n}}$	$\frac{1.22}{\sqrt{n}}$	$\frac{1.36}{\sqrt{n}}$	$\frac{1.52}{\sqrt{n}}$	$\frac{1.63}{\sqrt{n}}$

Source: Adapted by permission from Table 1 of Leslie H. Miller, Table of Percentage points of Kolmogrov statistics, *J. Am. Stat. Assoc.* 51 (1956), 111–121.

^aThis table gives the values of $D_{n,\alpha}^+$ and $D_{n,\alpha}$ for which $\alpha \geq P\{D_n^+ > D_{n,\alpha}^+\}$ and $\alpha \geq P\{D_n > D_{n,\alpha}\}$ for some selected values of n and α .

Table ST8. Critical Values of the Kolmogorov–Smirnov Test Statistic for Two Samples of Equal Size^a

One-Sided Test:											
$\alpha =$	0.10	0.05	0.025	0.01	0.005	$\alpha =$	0.10	0.05	0.025	0.01	0.005
Two-Sided Test:											
$\alpha =$	0.20	0.10	0.05	0.02	0.01	$\alpha =$	0.20	0.10	0.05	0.02	0.01
$n = 3$	2/3	2/3				$n = 20$	6/20	7/20	8/20	9/20	10/20
4	3/4	3/4	3/4			21	6/21	7/21	8/21	9/21	10/21
5	3/5	3/5	4/5	4/5	4/5	22	7/22	8/22	8/22	10/22	10/22
6	3/6	4/6	4/6	5/6	5/6	23	7/23	8/23	9/23	10/23	10/23
7	4/7	4/7	5/7	5/7	5/7	24	7/24	8/24	9/24	10/24	11/24
8	4/8	4/8	5/8	5/8	6/8	25	7/25	8/25	9/25	10/25	11/25
9	4/9	5/9	5/9	6/9	6/9	26	7/26	8/26	9/26	10/26	11/26
10	4/10	5/10	6/10	6/10	7/10	27	7/27	8/27	9/27	11/27	11/27
11	5/11	5/11	6/11	7/11	7/11	28	8/28	9/28	10/28	11/28	12/28
12	5/12	5/12	6/12	7/12	7/12	29	8/29	9/29	10/29	11/29	12/29
13	5/13	6/13	6/13	7/13	8/13	30	8/30	9/30	10/30	11/30	12/30
14	5/14	6/14	7/14	7/14	8/14	31	8/31	9/31	10/31	11/31	12/31
15	5/15	6/15	7/15	8/15	8/15	32	8/32	9/32	10/32	12/32	12/32
16	6/16	6/16	7/16	8/16	9/16	34	8/34	10/34	11/34	12/34	13/34
17	6/17	7/17	7/17	8/17	9/17	36	9/36	10/36	11/36	12/36	13/36
18	6/18	7/18	8/18	9/18	9/18	38	9/38	10/38	11/38	13/38	14/38
19	6/19	7/19	8/19	9/19	9/19	40	9/40	10/40	12/40	13/40	14/40
Approximation for $n > 40$:							$\frac{1.52}{\sqrt{n}}$	$\frac{1.73}{\sqrt{n}}$	$\frac{1.92}{\sqrt{n}}$	$\frac{2.15}{\sqrt{n}}$	$\frac{2.30}{\sqrt{n}}$

Source: Adapted by permission from Tables 2 and 3 of Z. W. Birnbaum and R. A. Hall, Small sample distributions for multisample statistics of the Smirnov type, *Ann. Math. Stat.* 31 (1960), 710–720.

^aThis table gives the values of $D_{n,n,\alpha}^+$ and $D_{n,n,\alpha}$ for which $\alpha \geq P\{D_{n,n}^+ > D_{n,n,\alpha}^+\}$ and $\alpha \geq P\{D_{n,n} > D_{n,n,\alpha}\}$ for some selected values of n and α .

Table ST9. Critical Values of the Kolmogorov–Smirnov Test Statistic for Two Samples of Unequal Size^a

One-Sided Test:	$\alpha =$	0.10	0.05	0.025	0.01	0.005
Two-Sided Test:	$\alpha =$	0.20	0.10	0.05	0.02	0.01
$N_1 = 1$	$N_2 = 9$	17/18				
	10	9/10				
$N_1 = 2$	$N_2 = 3$	5/6				
	4	3/4				
	5	4/5	4/5			
	6	5/6	5/6			
	7	5/7	6/7			
	8	3/4	7/8	7/8		
	9	7/9	8/9	8/9		
$N_1 = 3$	10	7/10	4/5	9/10		
	$N_2 = 4$	3/4	3/4			
	5	2/3	4/5	4/5		
	6	2/3	2/3	5/6		
	7	2/3	5/7	6/7	6/7	
	8	5/8	3/4	3/4	7/8	
	9	2/3	2/3	7/9	8/9	8/9
	10	3/5	7/10	4/5	9/10	9/10
	12	7/12	2/3	3/4	5/6	11/12
	$N_2 = 5$	3/5	3/4	4/5	4/5	
$N_1 = 4$	6	7/12	2/3	3/4	5/6	5/6
	7	17/28	5/7	3/4	6/7	6/7
	8	5/8	5/8	3/4	7/8	7/8
	9	5/9	2/3	3/4	7/9	8/9
	10	11/20	13/20	7/10	4/5	4/5
	12	7/12	2/3	2/3	3/4	5/6
	16	9/16	5/8	11/16	3/4	13/16
$N_1 = 5$	$N_2 = 6$	3/5	2/3	2/3	5/6	5/6
	7	4/7	23/35	5/7	29/35	6/7
	8	11/20	5/8	27/40	4/5	4/5
	9	5/9	3/5	31/45	7/9	4/5
	10	1/2	3/5	7/10	7/10	4/5
	15	8/15	3/5	2/3	11/15	11/15
	20	1/2	11/20	3/5	7/10	3/4

Table ST9. (Continued)

One-Sided Test:		$\alpha = 0.10$	0.05	0.025	0.01	0.005
Two-Sided Test:		$\alpha = 0.20$	0.10	0.05	0.02	0.01
$N_1 = 6$	$N_2 = 7$	23/42	4/7	29/42	5/7	5/6
		8 1/2	7/12	2/3	3/4	3/4
		9 1/2	5/9	2/3	13/18	7/9
		10 1/2	17/30	19/30	7/10	11/15
		12 1/2	7/12	7/12	2/3	3/4
		18 4/9	5/9	11/18	2/3	13/18
$N_1 = 7$	$N_2 = 8$	24 11/24	1/2	7/12	5/8	2/3
		27/56	33/56	5/8	41/56	3/4
		9 31/63	5/9	40/63	5/7	47/63
		10 33/70	39/70	43/70	7/10	5/7
		14 3/7	1/2	4/7	9/14	5/7
$N_1 = 8$	$N_2 = 9$	28 3/7	13/28	15/28	17/28	9/14
		4/9	13/24	5/8	2/3	3/4
		10 19/40	21/40	23/40	27/40	7/10
		12 11/24	1/2	7/12	5/8	2/3
		16 7/16	1/2	9/16	5/8	5/8
$N_1 = 9$	$N_2 = 10$	32 13/32	7/16	1/2	9/16	19/32
		7/15	1/2	26/45	2/3	31/45
		12 4/9	1/2	5/9	11/18	2/3
		15 19/45	22/45	8/15	3/5	29/45
		18 7/18	4/9	1/2	5/9	11/18
$N_1 = 10$	$N_2 = 15$	36 13/36	5/12	17/36	19/36	5/9
		2/5	7/15	1/2	17/30	19/30
		20 2/5	9/20	1/2	11/20	3/5
$N_1 = 12$	$N_2 = 40$	7/20	2/5	9/20	1/2	
		23/60	9/20	1/2	11/20	7/12
		16 3/8	7/16	23/48	13/24	7/12
		18 13/36	5/12	17/36	19/36	5/9
$N_1 = 15$	$N_2 = 20$	11/30	5/12	7/15	31/60	17/30
		7/20	2/5	13/30	29/60	31/60
$N_1 = 16$	$N_2 = 20$	27/80	31/80	17/40	19/40	41/80
Large-sample approximation		$1.07\sqrt{\frac{m+n}{mn}}$	$1.22\sqrt{\frac{m+n}{mn}}$	$1.36\sqrt{\frac{m+n}{mn}}$	$1.52\sqrt{\frac{m+n}{mn}}$	$1.63\sqrt{\frac{m+n}{mn}}$

Source: Adapted by permission from F. J. Massey, Distribution table for the deviation between two sample cumulatives, *Ann. Math. Stat.* 23 (1952), 435–441.

^aThis table gives the values of $D_{m,n,\alpha}^+$ and $D_{m,n,\alpha}$ for which $\alpha \geq P\{D_{m,n}^+ > D_{m,n,\alpha}^+\}$ and $\alpha \geq P\{D_{m,n} > D_{m,n,\alpha}\}$ for some selected values of N_1 = smaller sample size, N_2 = larger sample size, and α .

Table ST10. Critical Values of the Wilcoxon Signed-Ranks Test Statistic^a

n	α			
	0.01	0.025	0.05	0.10
3	6	6	6	6
4	10	10	10	9
5	15	15	14	12
6	21	20	18	17
7	27	25	24	22
8	34	32	30	27
9	41	39	36	34
10	49	46	44	40
11	58	55	52	48
12	67	64	60	56
13	78	73	69	64
14	89	84	79	73
15	100	94	89	83
16	112	106	100	93
17	125	118	111	104
18	138	130	123	115
19	152	143	136	127
20	166	157	149	140

Source: Adapted by permission from Table 1 of R. L. McCornack, Extended tables of the Wilcoxon matched pairs signed-rank statistics, *J. Am. Stat. Assoc.* 60 (1965), 864–871.

^aThis table gives values of t_α for which $P\{T^+ > t_\alpha\} \leq \alpha$ for selected values of n and α . Critical values in the lower tail may be obtained by symmetry from the equation $t_{1-\alpha} = n(n+1)/2 - t_\alpha$.

Table ST11. Critical Values of the Mann–Whitney–Wilcoxon Test Statistic^a

<i>m</i>	α	<i>n</i>								
		2	3	4	5	6	7	8	9	10
2	0.01	4	6	8	10	12	14	16	18	20
	0.025	4	6	8	10	12	14	15	17	19
	0.05	4	6	8	9	11	13	14	16	18
	0.10	4	5	7	8	10	12	13	15	16
3	0.01		9	12	15	18	20	20	25	28
	0.025		9	12	14	16	19	21	24	26
	0.05		8	11	13	15	18	20	22	25
	0.10		7	10	12	14	16	18	21	23
4	0.01			16	19	22	26	29	32	36
	0.025			15	18	21	24	27	31	34
	0.05			14	17	20	23	26	29	32
	0.10			12	15	18	21	24	26	29
5	0.01				23	27	31	35	39	43
	0.025				22	26	29	33	37	41
	0.05				20	24	28	31	35	38
	0.10				19	22	26	29	32	36
6	0.01					32	37	41	46	51
	0.025					30	35	39	43	48
	0.05					28	33	37	41	45
	0.10					26	30	34	38	42
7	0.01						42	48	53	58
	0.025						40	45	50	55
	0.05						37	42	47	52
	0.10						35	39	44	48
8	0.01							54	60	66
	0.025							50	56	62
	0.05							48	53	59
	0.10							44	49	55
9	0.01								66	73
	0.025								63	69
	0.05								59	65
	0.10								55	61
10	0.01									80
	0.025									76
	0.05									72
	0.10									67

Source: Adapted by permission from Table 1 of L. R. Verdooren, Extended tables of critical values for Wilcoxon's test statistic, *Biometrika* 50 (1963), 177–186, with the kind permission of Professor E. S. Pearson, the author, and the *Biometrika* Trustees.

^aThis table gives values of u_α for which $P\{U > u_\alpha\} \leq \alpha$ for some selected values of m , n , and α . Critical values in the lower tail may be obtained by symmetry from the equation $u_{1-\alpha} = mn - u_\alpha$.

Table ST12. Critical Points of Kendall's Tau Test Statistic^a

n	α			
	0.100	0.050	0.025	0.01
3	3	3	3	3
4	4	4	6	6
5	6	6	8	8
6	7	9	11	11
7	9	11	13	15
8	10	14	16	18
9	12	16	18	22
10	15	19	21	25

Source: Adapted by permission from Table 1, p. 173, of M. G. Kendall, *Rank Correlation Methods*, 3rd ed., Griffin, London, 1962. For values of $n \geq 11$, see W. J. Conover, *Practical Nonparametric Statistics*, John Wiley, New York, 1971, p. 390.

^aThis table gives the values of S_α for which $P\{S > S_\alpha\} \leq \alpha$, where $S = \binom{n}{2}T$, for some selected values of α and n . Values in the lower tail may be obtained by symmetry, $S_{1-\alpha} = -S_\alpha$.

Table ST13. Critical Values of Spearman's Rank Correlation Statistic^a

n	α			
	0.01	0.025	0.05	0.10
3	1.000	1.000	1.000	1.000
4	1.000	1.000	0.800	0.800
5	0.900	0.900	0.800	0.700
6	0.886	0.829	0.771	0.600
7	0.857	0.750	0.679	0.536
8	0.810	0.714	0.619	0.500
9	0.767	0.667	0.583	0.467
10	0.721	0.636	0.552	0.442

Source: Adapted by permission from Table 2, pp. 174–175, of M. G. Kendall, *Rank Correlation Methods*, 3rd ed., Griffin, London, 1962. For values of $n \geq 11$, see W. J. Conover, *Practical Nonparametric Statistics*, John Wiley, New York, 1971, p. 391.

^aThis table gives the values of R_α for which $P\{R > R_\alpha\} \leq \alpha$ for some selected values of n and α . Critical values in the lower tail may be obtained by symmetry, $R_{1-\alpha} = -R_\alpha$.